PROBLEM STATEMENT : Task 1: Create a Silicon and Germanium PN with P-type doping 1e18 /cm^3 and n-type doping 1e16 /cm^3.

Step 1: Wafer Initialization

Purpose: To create the initial silicon substrate with background doping.

Physical Change: A silicon wafer is generated with uniform n-type (phosphorus) doping.

Step 2: Oxide Growth

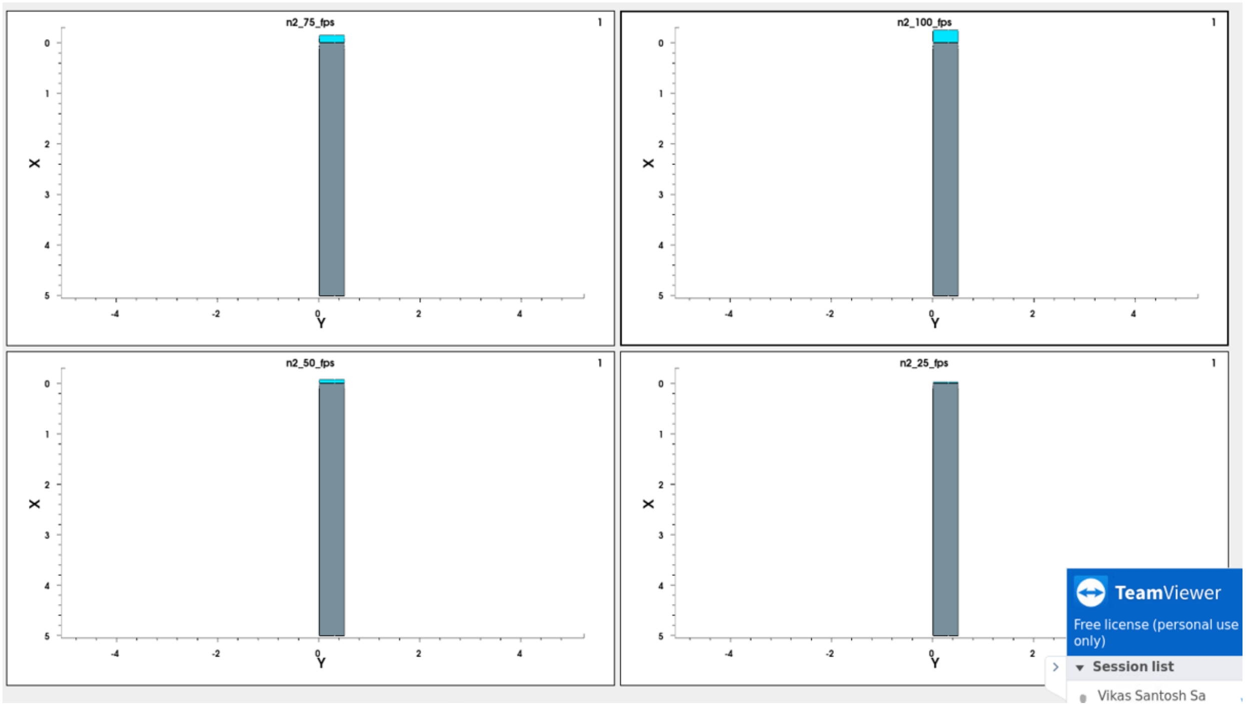
Purpose: To form a high-quality silicon dioxide (SiO_2) layer that acts as a gate dielectric, mask, or protective barrier.

Physical Change: The wafer is exposed to oxygen or steam at high temperature (typically 900–1200°C), causing the exposed silicon surface itself to chemically react and convert into SiO_2 (thermal oxidation). This consumes part of the underlying silicon, producing a dense, defect-free oxide that is tightly bonded to the substrate.

Step 3: Oxide Deposition

Purpose: To add an insulating or masking oxide layer without consuming the substrate silicon, often used for field oxide, interlayer dielectrics, or spacers.

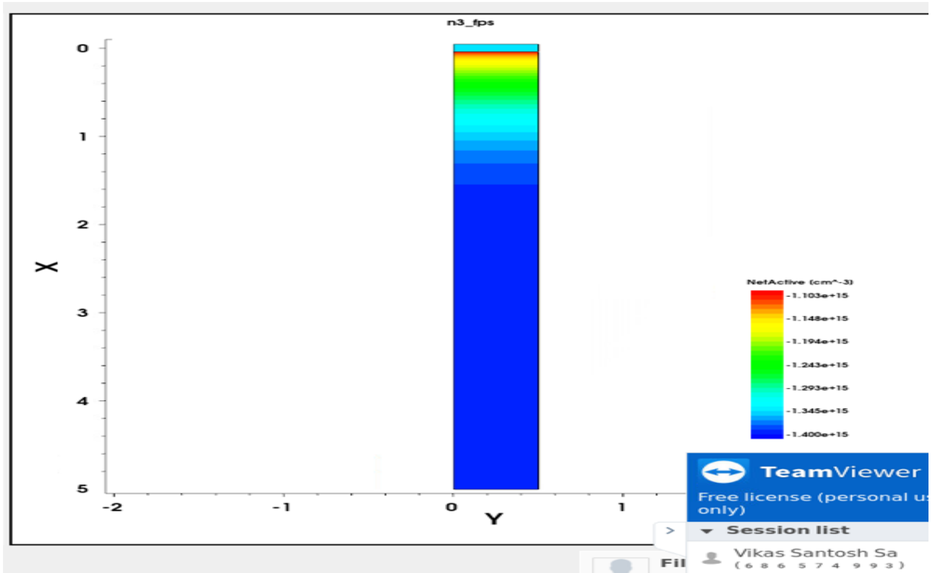
Physical Change: SiO_2 (or another dielectric) is deposited on top of the existing wafer surface using techniques like Chemical Vapor Deposition (CVD), rather than growing from the silicon itself. The layer forms additively on the surface, so the original silicon underneath remains unconsumed.



Step 4: Implantation

Purpose: To introduce dopant atoms (e.g., boron for p-type, phosphorus/arsenic for n-type) into specific regions of the wafer to define source, drain, well, or channel regions.

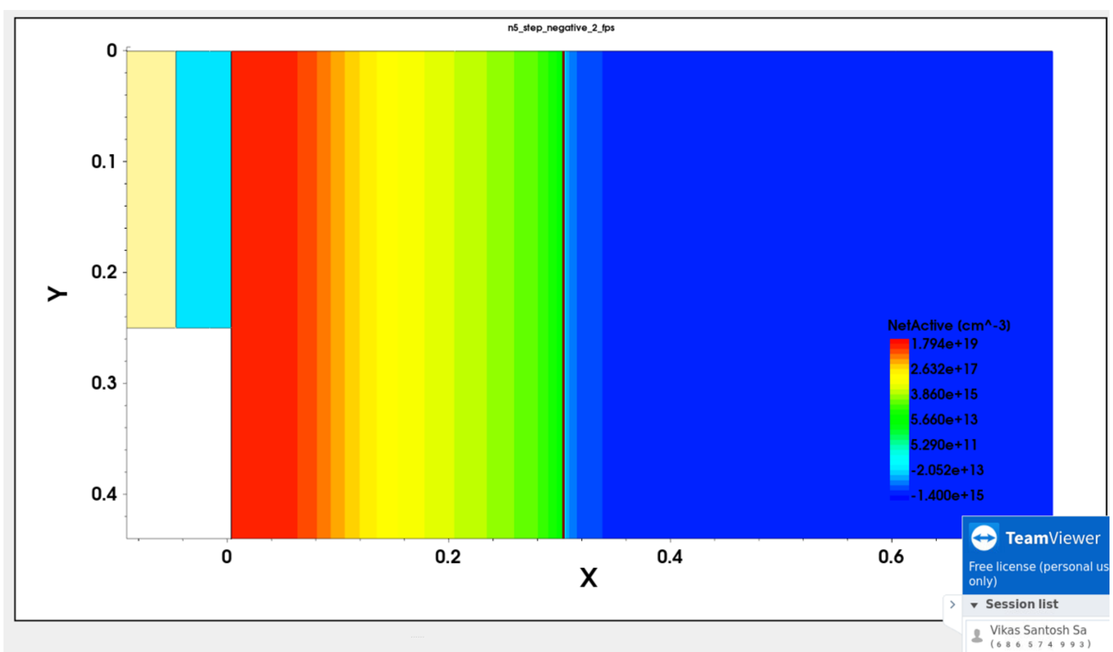
Physical Change: Ions of the dopant species are accelerated and fired into the silicon lattice using an electric field. This damages the crystal structure locally and embeds the dopant atoms at a controlled depth and concentration, determined by the ion energy and dose. A subsequent anneal step (not listed here) would repair the lattice and electrically activate the dopants.



Step 5: Etching

Purpose: To selectively remove material (oxide, polysilicon, or other layers) to define patterns such as contact windows, gate structures, or isolation trenches.

Physical Change: A new oxide layer grows on the exposed silicon surfaces around the gate.



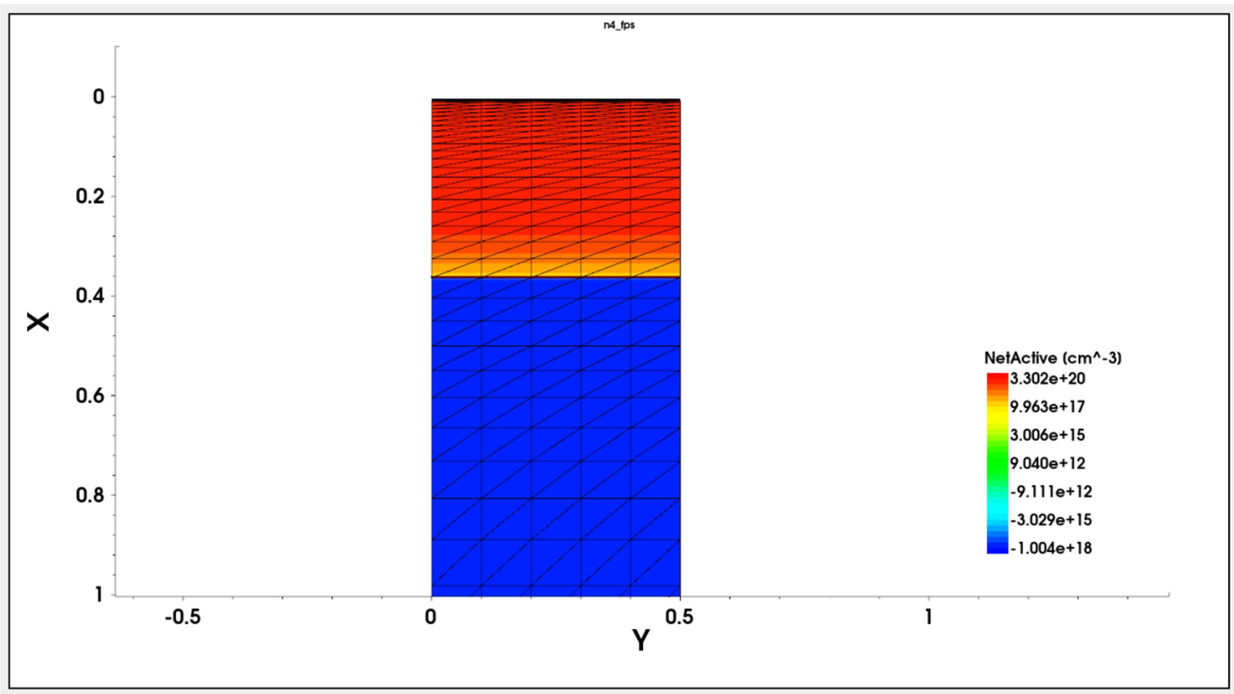
⇨ Meshing

- ⇨ Meshing is the process of dividing the simulation domain (the wafer/device structure) into a finite number of small, discrete elements or grid points, at which the simulator numerically solves the governing semiconductor equations (Poisson's equation, carrier continuity equations, etc.). Rather than solving these equations continuously across the entire device, meshing converts the structure into a **discrete grid** so that a numerical (finite-element or finite-difference) solution becomes computationally feasible.
- ⇨ The mesh is typically **non-uniform** — finer (denser) near critical regions such as the junction, oxide-silicon interfaces, or contact areas where physical quantities (doping concentration, electric field, carrier density) change rapidly, and coarser in regions like the bulk substrate where values vary slowly.

Importance of Meshing

1. Enables numerical solution of device physics – Semiconductor device equations generally have no closed-form analytical solution for realistic structures; meshing converts the continuous problem into a solvable system of discrete equations.
2. Determines simulation accuracy – A mesh that is too coarse near the PN junction can miss sharp variations in doping profile or electric field, leading to inaccurate predictions of junction behavior, depletion width, or breakdown voltage.
3. Affects computational efficiency – A finer mesh gives higher accuracy but increases computation time and memory usage; an appropriately graded mesh balances accuracy where it matters (junction region) against efficiency elsewhere (bulk regions).
4. Captures steep gradients correctly – Near the metallurgical junction, doping concentration and electric field change over very short distances. Dense meshing in this region ensures these steep gradients are resolved properly instead of being averaged out or missed.

5. Impacts convergence of the simulation – Poorly structured meshes (e.g., abrupt size changes between adjacent elements) can cause numerical **instability or failure to converge during the solving process, so mesh quality directly affects whether the simulation runs successfully at all.**



PROBLEM STATEMENT : With the given template in the resources of Synopsys Sentaurus TCAD (Applications_Library/GettingStarted/sprocess/2DGS), create 180 nm 2D nMOS without using reflection in the last step.

Step 1: Wafer Initialization

Purpose: To create the initial silicon substrate with background doping.

Physical Change: A silicon wafer is generated with uniform n-type (phosphorus) doping.

Step 2: Well Implantation & Annealing

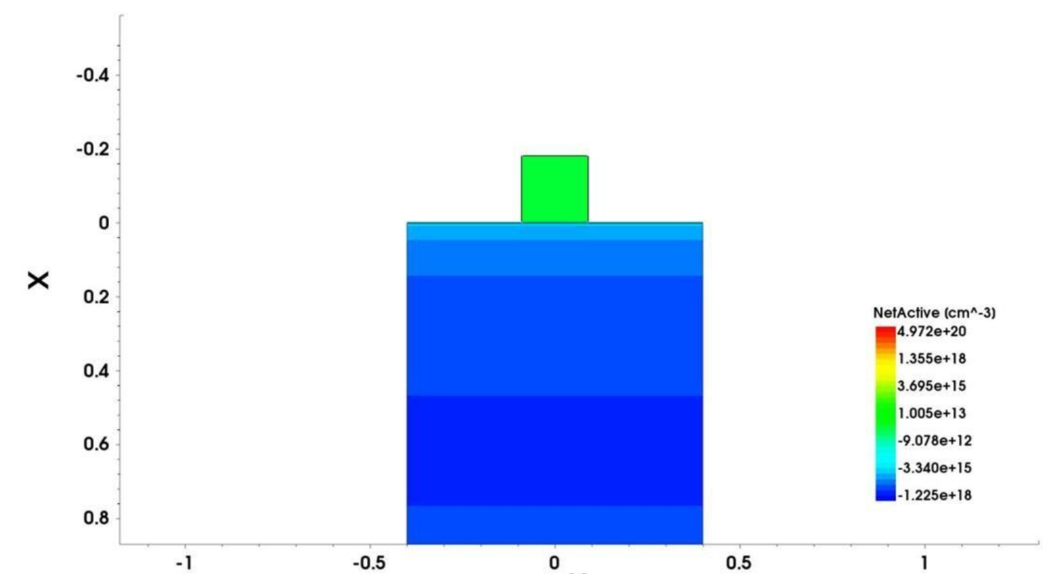
Purpose: To form the p-well region for the nMOS device.

Physical Change: Boron implantation creates a p-type region, and annealing activates and redistributes the dopants while growing a thin oxide layer.

Step 3: Polysilicon Deposition & Gate Patterning

Purpose: To form the MOS gate electrode.

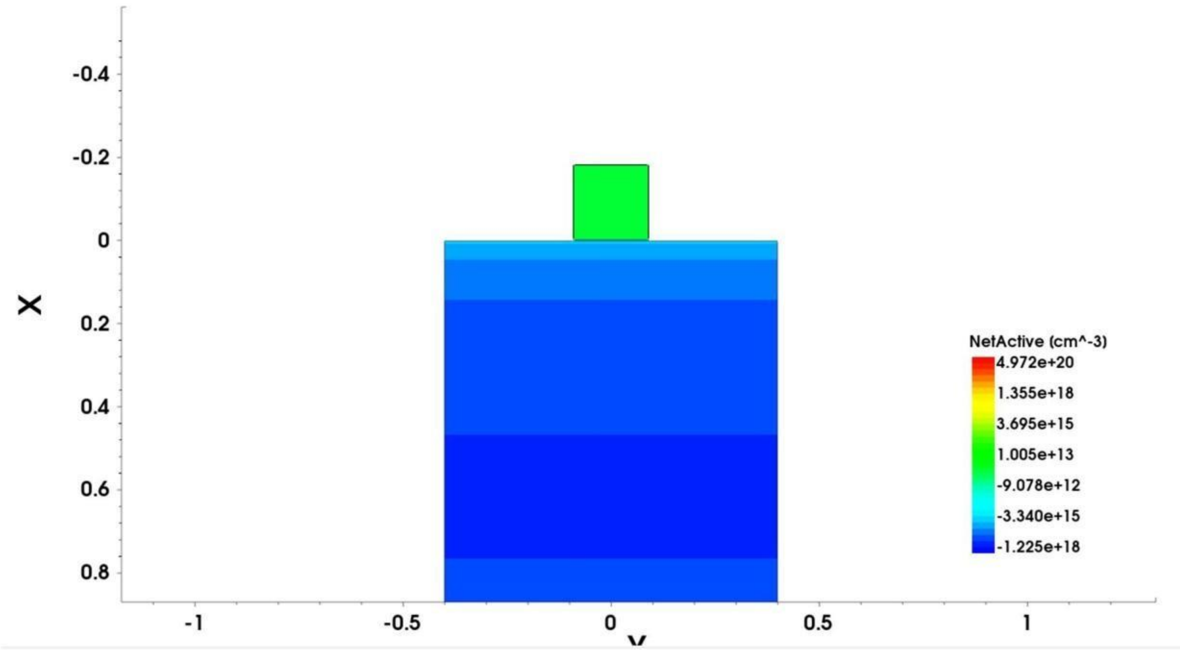
Physical Change: A polysilicon layer is deposited and etched, leaving the patterned gate at the center.



Step 4: Gate Oxide Etching

Purpose: To remove unwanted oxide outside the gate region.

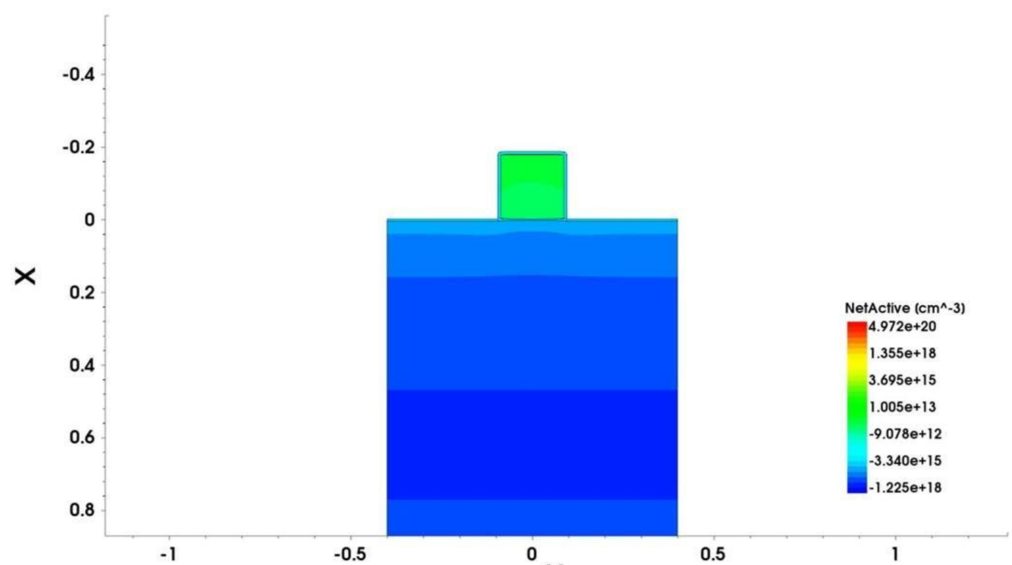
Physical Change: Oxide is etched away from exposed regions while oxide beneath the gate remains.



Step 5: Thermal Oxidation

Purpose: To grow a protective oxide layer on exposed silicon.

Physical Change: A new oxide layer grows on the exposed silicon surfaces around the gate.



Step 6: Mesh Refinement

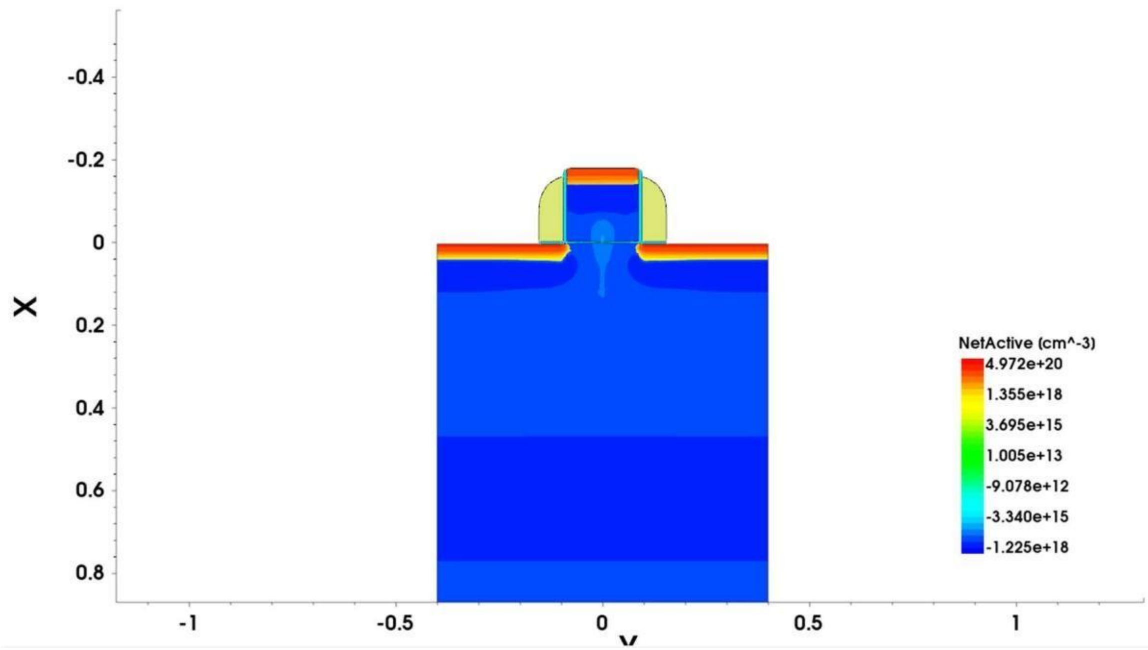
Purpose: To improve simulation accuracy near the device region.

Physical Change: The physical structure remains unchanged; only the computational mesh becomes finer around the channel.

Step 7: LDD Implantation & Annealing Step 8: Spacer Forma on

Purpose: To create sidewall spacers on the gate.

Physical Change: Nitride is deposited and anisotropically etched, leaving spacers on both sides of the gate.



Step 9: Mesh Refinement

Purpose: To improve simulation accuracy near the source, gate and drain region.

Physical Change: The physical structure remains unchanged; only the computational mesh becomes finer around the channel.

Step 10: Source/Drain Forma on & Contact Forma on (Final Structure)

Purpose: To complete the nMOS transistor and provide metal contacts.

Physical Change: Heavy arsenic implantation forms n^+ source/drain regions, aluminum contacts are patterned, resulting in the completed
180 nm 2D nMOS